

Fourier Transform - Properties

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	Signal	Fourier transform (unitary, k domain)	Remarks
	$f(x) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(k) e^{ikx} dk$	$F(k) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$	
1	$a \cdot f(x) + b \cdot h(x)$	$a \cdot F(k) + b \cdot H(k)$	Linearity
2	$f(x - a)$	$e^{-iak} F(k)$	Shift in the x domain
3	$e^{iax} f(x)$	$F(k - a)$	Shift in the k domain
4	$f(ax)$	$\frac{1}{ a } F\left(\frac{k}{a}\right)$	If $ a $ is large, then $f(ax)$ is concentrated around 0 and $\frac{1}{ a } F\left(\frac{k}{a}\right)$ spreads out
5	$F(x)$	$f(-k)$	Duality property of the Fourier transform
6	$\frac{d^n f(x)}{dx^n}$	$(ik)^n F(k)$	Generalized derivative property of the Fourier transform
7	$x^n f(x)$	$i^n \frac{d^n F(k)}{dk^n}$	This is the dual to 6
8	$\int_{-\infty}^x f(y) dy$	$\frac{1}{\sqrt{2\pi}} \left(\frac{F(k)}{ik} + \pi F(0) \delta(k) \right)$	Fourier transform of integral
9	$(f * h)(x)$	$\sqrt{2\pi} F(k) H(k)$	$f * h$ denotes the convolution of f and h — this rule is the convolution theorem
10	$f(x)h(x)$	$\frac{(F*H)(k)}{\sqrt{2\pi}}$	This is the dual of 9
11	For a purely real even function $f(x)$	$F(k)$ is a purely real even function	
12	For a purely real odd function $f(x)$	$F(k)$ is a purely imaginary odd function	

Table 1: Fourier Transform Properties